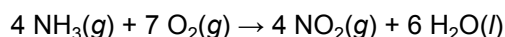


When ammonia burns in air, nitrogen dioxide and water are produced according to the reaction below:



At 0 °C and 1 atm pressure, burning 44.8 L of ammonia requires:

- ☐ A. 6/7 moles O₂. [3%]
☐ B. 8/7 moles O₂. [6%]
☐ C. 7/4 moles O₂. [25%]
☒ D. 7/2 moles O₂. [63%]

Correct

64% answered correctly

Explanation:

$$4 \text{NH}_3(g) + 7 \text{O}_2(g) \longrightarrow 4 \text{NO}_2(g) + 6 \text{H}_2\text{O}(l)$$

Volume reacting	Molar volume of gas at STP	Mole ratio of reaction	Moles O ₂ consumed
44.8 L-NH ₃ -gas	$\times \frac{1 \text{ mol-NH}_3}{22.4 \text{ L-NH}_3\text{-gas}}$	$\times \frac{7 \text{ mol O}_2}{4 \text{ mol-NH}_3}$	$= \frac{7}{2} \text{ mol O}_2$

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Under conditions of **standard temperature and pressure (STP)**, which is defined as **0 °C** and **1 atm** of pressure, a gas occupies a volume of **22.4 L for every 1 mole** of gas molecules. The moles of a gas present in a given volume at STP can be determined by using this relationship. In a reaction, the moles of gas can be related to the moles of other chemical species by applying stoichiometric **mole ratios** obtained from the **coefficients** of a **balanced reaction equation**.

Because the ammonia in the given reaction was burned at 0 °C and 1 atm (STP), the number of moles of NH₃ present in 44.8 L of gas is given by:

$$44.8 \text{ L} \times \frac{1 \text{ mol-NH}_3}{22.4 \text{ L-NH}_3\text{-gas}} = 2.00 \text{ mol-NH}_3$$

According to the balanced equation, 7 moles of O₂ are required for every 4 moles of NH₃ that are burned. Using this molar ratio, the required number of moles of O₂ can be found:

$$2.00 \text{ mol-NH}_3 \times \frac{7 \text{ mol O}_2}{4 \text{ mol-NH}_3} = 3.50 \text{ mol O}_2$$

(Choice A) This is the stoichiometric ratio of the moles of H₂O to the moles of O₂. The molar ratio between H₂O and O₂ cannot be used for the molar proportions between NH₃ and O₂.

(Choice B) This value is obtained if the stoichiometric ratio of the moles of NH₃ burned to the moles of O₂ consumed is mistakenly inverted during the second step of the calculation.

(Choice C) This is the correct stoichiometric ratio between the moles of NH₃ and the moles of O₂, but this ratio alone does not account for the actual number of moles of ammonia burned during the reaction.

Educational objective:

Under conditions of standard temperature and pressure (STP), which is defined as 0 °C and 1 atm of pressure, 1 mole of gas occupies a volume of 22.4 L. Mole ratios from balanced chemical reactions can be used to relate the moles of a given gas to the moles of other reaction species.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws